AWS SAA-C03 Study Notes: Data Analytics & Big Data

This guide covers the key Data Analytics services and architectural patterns frequently tested in the AWS Certified Solutions Architect - Associate (SAA-C03) exam.

1. Key Services Overview

Amazon Athena

- **What it is:** Serverless, interactive query service.
- **Key Function:** Analyze data directly in **Amazon S3** using standard **SQL**.
- **Performance & Cost Optimization (Exam Critical):**
  - **Columnar Data:** Use **Parquet** or **ORC** formats. Athena scans *only* the columns needed (saves money/time) vs. CSV/JSON where it scans the whole file.
  - **Compression:** Compress data (Snappy, Zlib, GZIP) to reduce scan size.
  - **Partitioning:** Organize S3 folders by date/category (e.g., `s3://bucket/year=2024/month=01/`). Athena scans only the relevant folder ("partition pruning").
- **Federated Query:**
  - Allows running SQL queries across data stored in relational, non-relational, object, and custom data sources (e.g., DynamoDB, RDS, Redshift) *without* moving data to S3.
  - Uses **Data Source Connectors** running on **AWS Lambda**.
- **Exam Keywords:** Serverless, Ad-hoc queries, S3, SQL, Parquet/ORC, "no infrastructure to manage."

Amazon Redshift

- **What it is:** Petabyte-scale, fully managed **Data Warehouse** (OLAP).
- **Architecture:**
  - **Leader Node:** Handles client connections (JDBC/ODBC), parses SQL, creates execution plans, and aggregates results. *You only interact with this node.*
  - **Compute Nodes:** Execute the queries in parallel and store the data. Data is distributed across "slices" in these nodes.
  - **Redshift Spectrum:** A serverless feature that lets Redshift query data directly in S3 (using the Leader Node) without loading it. Ideal for "cold" data or infrequent queries.
- **Data Loading & Ingestion:**
  - **COPY Command:** The fastest way to load batch data from S3.
  - **Kinesis Data Firehose:** For **streaming ingestion** directly into Redshift (near real-time).

    - **JDBC/ODBC Driver:** Standard drivers for connecting BI tools (Tableau, QuickSight) or applications.
- **Security & Networking:**
    - **Enhanced VPC Routing:** Forces all COPY and UNLOAD traffic between Redshift and S3 to go through your VPC (using VPC Endpoints), bypassing the public internet. Critical for strict compliance.
    - **Disaster Recovery (DR):**
        - **Snapshots:** Automated or manual point-in-time backups stored in S3.
        - **Cross-Region Replication:** Configure Redshift to automatically copy snapshots to another region for DR.

Amazon OpenSearch Service (formerly Elasticsearch)

- **What it is:** Managed service for search, analytics, and log monitoring.
- **Key Function:** Operational analytics (application monitoring), full-text search, and log analysis.
- **Exam Keywords:** Search, Indexing, Kibana (dashboards), Log analytics, "fuzzy search."
- **Integration:** Often acts as the destination for logs coming from Kinesis Data Firehose or CloudWatch Logs.

Amazon EMR (Elastic MapReduce)

- **What it is:** Managed cluster platform for big data frameworks (Hadoop, Spark, Hive, Presto).
- **Cluster Architecture (Node Types):**
    - **Master Node:** Manages the cluster (YARN ResourceManager, HDFS NameNode). Coordinates the distribution of data and tasks.
    - **Core Node:** Runs tasks *and* stores data in HDFS. **Do not** terminate these easily (risk data loss).
    - **Task Node:** Runs tasks *only* (no data storage). You can add/remove these to scale compute power. **Perfect usage for Spot Instances**.
- **Exam Keywords:** Hadoop, Spark, Big Data migration, Spot Instances (Task nodes).

Amazon QuickSight

- **What it is:** Serverless Business Intelligence (BI) service.
- **Key Function:** Create and publish interactive dashboards and visualizations.
- **User & Group Management:**
    - **Standard Edition:** Manage users via IAM or email invites.
    - **Enterprise Edition:** Supports **Active Directory (AD)** integration and Federated Login.
    - **Row-Level Security (RLS):** Restrict access to specific data rows based on User or Group (e.g., "Sales group sees only Sales data").

- **Exam Keywords:** BI, Visualizations, Dashboards, Machine Learning Insights, "pay-per-session."

AWS Glue

- **What it is:** Serverless Data Integration (ETL - Extract, Transform, Load) service.
- **Key Function:** Prepare and load data for analytics.
- **Components:**
  - **Data Catalog:** Central repository for metadata (schema definitions). Required for Athena and Redshift Spectrum.
  - **Crawler:** Automatically scans S3 data to infer schema and update the Data Catalog.
  - **Glue Jobs:** Python/Scala scripts to transform data (e.g., convert CSV to Parquet).
- **Exam Keywords:** ETL, Serverless, Data Catalog, Schema discovery.

AWS Lake Formation

- **What it is:** Service to set up, secure, and manage a Data Lake.
- **Key Function:** Simplifies security and governance. It sits *on top* of Glue and S3.
- **Exam Keywords:** "Fine-grained access control" (row-level/column-level security), "Blueprint," Governance, "secure data lake in days."

Amazon Managed Service for Apache Flink

- **Note:** Formerly *Kinesis Data Analytics*.
- **What it is:** Fully managed service for real-time stream processing.
- **Key Function:** Run stateful applications to process data streams using Java, Scala, Python, or SQL.
- **Exam Keywords:** Real-time analytics, Stateful processing, Windowing (e.g., "calculate average every 5 minutes"), Flink.

Amazon MSK (Managed Streaming for Apache Kafka)

- **What it is:** Fully managed Apache Kafka service.
- **Key Function:** Ingest and process streaming data using Kafka.
- **Exam Keywords:** "Migrate existing Kafka cluster," Open-source compatibility, High throughput.

2. Common Big Data Ingestion Pipelines (Architectural Patterns)

The SAA-C03 exam loves to test how these services connect. Here are the three most common patterns:

Pattern A: The "Modern Data Lake" (Batch Processing)

Scenario: Ingest raw data, transform it, and analyze it using SQL.

1. **Ingest:** Data lands in **S3** (Raw Bucket).
2. **Catalog: Glue Crawler** scans S3 and updates **Glue Data Catalog**.
3. **Process (ETL): Glue Jobs** (or EMR) clean data and convert to Parquet.
4. **Store:** Transformed data saved to **S3** (Processed Bucket) or loaded into **Redshift**.
5. **Analyze:**
   - **Athena** queries S3 directly (Ad-hoc).
   - **Redshift** queries data warehouse (Complex BI).
   - **QuickSight** visualizes the results.

Pattern B: The "Real-Time Streaming" Pipeline

Scenario: Analyze clickstream data or IoT sensors in real-time.

1. **Ingest: Kinesis Data Streams** (or MSK) captures high-velocity data.
2. **Process: Managed Service for Apache Flink** processes the stream (e.g., aggregates metrics).
3. **Store/Deliver:**
   - **Kinesis Data Firehose** delivers the processed data to **S3**, **Redshift**, or **OpenSearch**.
4. **Analyze:** Real-time dashboards in **OpenSearch** (Kibana) or **QuickSight**.

Pattern C: Log Analytics

Scenario: Centralized analysis of application logs.

1. **Source:** CloudWatch Logs or EC2 logs.
2. **Stream: Kinesis Data Firehose**.
3. **Destination: Amazon OpenSearch Service**.
4. **Visualize: Kibana** (built into OpenSearch) for searching and visualizing error logs.

3. Quick Comparison Cheat Sheet

Feature	Athena	Redshift	EMR
Primary Use	Ad-hoc S3 queries	Data Warehousing / BI	Big Data Processing (Spark/Hadoop)
Infrastructure	Serverless	Clusters (Managed)	Clusters (Managed)
Pricing	Per TB scanned	Per Node / Hour	Per Instance / Hour
Speed	Seconds/Minutes	Milliseconds/Seconds	Minutes/Hours (Batch)
Exam Trigger	"Serverless SQL on S3"	"Complex joins", "Warehouse"	"Apache Spark", "Hadoop migration"

Feature	Kinesis Data Streams	Kinesis Data Firehose
Real-time?	Yes (Real-time)	Near Real-time (60s buffer)
Function	Ingest custom streams	**Load** data into destinations
Destinations	Lambda, Flink, EC2	S3, Redshift, OpenSearch, Splunk
Management	Manual scaling (shards)	Fully Serverless (Auto-scaling)